Homework Questions: Authentication & Cryptography

Question 1

Topic: Password Security – Salting & Rainbow Tables
Explain what a cryptographic **salt** is and how it prevents an attacker from successfully using a precomputed **rainbow table** to crack password hashes.

Question 2

Topic: Biometrics – Sensitivity vs. Specificity
Biometric authentication systems face a fundamental trade-off between **sensitivity** (true positive rate) and **specificity** (true negative rate).
Imagine you are designing a biometric door lock for a **high-security military base**. Which metric would you prioritize and why? What would be the real-world consequence of getting this trade-off wrong?

Question 3

Topic: Message Authentication – Hash Functions vs. HMAC
A plain cryptographic hash function such as SHA-256 cannot be used directly as a **Message Authentication Code (MAC)** .

- Why is this the case?
- How does **HMAC** (Hash-based Message Authentication Code) address this limitation to provide secure message authentication?

Question 4

Topic: Public Key Cryptography – MITM Attacks & Certificates

- Describe how a **man-in-the-middle (MITM) attack** works against a public key exchange protocol such as Diffie-Hellman.
- Then explain how **digital certificates**, issued by a **Certificate Authority (CA)** , prevent an attacker from successfully executing such an attack.

Question 5

Topic: Multi-Factor Authentication (MFA)
A bank implements two-factor authentication (2FA) by requiring customers to enter:

1. A **password** (something you know)
2. A **one-time code** sent via SMS text message (something you have)

- Which of the three standard authentication factors is missing from this

system?
- What specific additional factor would the bank need to require in order to achieve **true three-factor authentication (3FA)** ?

Question 6

Topic: Symmetric vs. Asymmetric Encryption
Compare and contrast **symmetric encryption** and **asymmetric (public-key) encryption**.
Your answer should include:

- At least **two advantages** of each approach
- At least **two disadvantages** of each approach
- One **real-world scenario** where you would choose symmetric encryption over asymmetric encryption, and one scenario where you would choose asymmetric encryption instead.

Submission Instructions

- Write your answers clearly and concisely. Use examples where appropriate.
- Include your name and student ID at the top of your submission.
- Submit your completed homework as a **PDF file**.